

Master Thesis Presentation

- Good afternoon, my name is Heykel Khadhraoui. I developed a surface grinding controller for a seven-joint robot.
- Normal pressing and compliant rotation allow contact to move the tool towards alignment. A separate experiment tests how the arm responds to disturbances.

Motivation

- Grinding needs the tool face aligned with the physical surface. The configured normal comes from calibration against the plate, with residual uncertainty.
- With a tilted tool, normal pressing creates a contact moment. Low rotational stiffness lets this moment turn the tool towards alignment.
- Cartesian impedance makes position and rotation compliant. The desired orientation stays fixed while the rotational spring and damper remain active.

Contact demonstration

- The video shows pickup, approach, contact, and the hold before grinding.
- During the hold, I change the surface orientation. Normal pressing continues as contact moments rotate the tool.
- The desired orientation stays fixed. This is a qualitative demonstration of compliant contact.

Overview

- I introduce the motivation and Cartesian controller, then the contact experiments and results.
- The separate null-space controller, experiment and results follow together. I finish with the conclusion and future work.

Cartesian pose

- The end effector (EE) is the tool or gripper at the end of the arm. Its Cartesian pose describes position and rotation relative to the robot base frame.
- Position uses x , y , and z , while roll, pitch, and yaw illustrate rotation. Together they give six degrees of freedom.
- Forward kinematics calculates the pose from all seven joint angles, with $p = p(q)$ for position. The controller uses a rotation matrix for orientation.

Cartesian impedance controller

- Positional error points from measured to desired position. Rotational error uses a correction angle ϕ and unit axis u .
- The sketch compares measured and desired tool orientations at one position. The axis points out of the page.
- Stiffness produces force and moment. Damping opposes velocity differences.

- Force and moment form a wrench. Its six-by-six stiffness and damping matrices have three-by-three blocks, with zero off-diagonal blocks here.
- Before the point shift, the stiffness and damping matrices are diagonal in the surface frame.

Real-time control

- The robot receives joint torques. We generate the Cartesian contribution from the wrench using the Jacobian transpose:

$$\tau_{\text{cart}} = J^{\top}(q)F$$

- The Jacobian maps joint velocities to the EE's linear and angular velocities. Model compensation and enabled null-space torques complete the command.
- The measured pose feeds back into the control loop. The C++ controller uses libfranka and updates every millisecond.

Translation–rotation coupling

- The tool centre point (TCP) is the controlled reference point attached to the EE. The centre of compliance (CoC) at p_c is a virtual reference for stiffness and damping.
- Moving the CoC changes how position and rotation errors generate force and moment. Its displacement from the TCP is $r_c = p_c - p_{\text{TCP}}$.
- The full matrix equation uses the same force–moment column and position–rotation error column as the Cartesian controller. The $A^{\top}$ and A transformations shift both stiffness and damping to the TCP.
- The underbraces identify the TCP stiffness and damping matrices. Their off-diagonal blocks couple translation and rotation.
- The error transforms as $e_c = Ae$, and the wrench transforms as $F_{\text{TCP}} = A^{\top}F_c$. These point-shift relations use the small-rotation model.

Centre of compliance (CoC)

- Shifting the CoC can change both the commanded force and moment. The shifted controller's force acts as if applied at p_c :
- The added moment can support or oppose alignment. For the same force, reversing the displacement reverses this contribution.
- With the CoC at the TCP, the added contribution vanishes. The rotational spring and damper still produce m_R .

Contact experiment

- The process goes from tool orientation and surface approach to contact establishment, then grinding. These experiments evaluate the five-second contact-establishment phase.
- Low normal stiffness allows compliant pressing, while higher tangential stiffness resists sideways movement. Rotational stiffness is higher about the normal and lower about the surface tangents.

- The position reference advances into the surface to generate pressing force. The desired orientation stays at its entry value, so contact moments drive the rotation.

Angular quantities

- The red arrow is the inward calibrated surface normal. The green and blue arrows show the tool normals calculated from measured EE orientation at contact entry and end.
- Both angle arcs start from the red reference. The entry angle is the measured angular offset, and the angular error is the remaining difference about the first surface tangent.
- Smaller magnitude means less error in this component. Calibration and mount errors affect the estimate of physical alignment.

Normal-force plausibility assessment

- I manually push the tool about 20 millimetres along the surface normal, with the CoC at the TCP. This checks the spring-force prediction with the desired pose fixed.
- The stiffness is 1000 newtons per metre and the measured position-error change is -19.65 millimetres. Their product gives the spring-force prediction:

$$F_{n,qs} = (1000 \text{ N/m}) \cdot (-19.65 \times 10^{-3} \text{ m}) = -19.65 \text{ N}$$

- In the shaded stationary interval, the commanded mean is -19.64 newtons, only 0.05 percent from the calculation. The robot's model estimate is -22.29 newtons, 13.50 percent from the command.

Moment plausibility assessment

- I manually rotate the tool about surface tangent one while keeping the normal push. This checks the spring-moment prediction with the desired pose fixed and CoC at the TCP.
- The measured rotational error changes by 6.57° relative to the loaded baseline. With a stiffness of 15 newton metres per radian, converting this change to radians gives:

$$M_{t1,qs} = (15 \text{ N m/rad}) \cdot (6.57 \cdot \frac{\pi}{180} \text{ rad}) \approx 1.72 \text{ N m}$$

- In the shaded stationary interval, the commanded mean matches the calculated 1.72 newton metres. The robot's model estimate is 1.85 newton metres, 7.75 percent from the command.

Commanded and estimated wrench

- A separate contact-establishment trial compares commanded force and moment with the robot's model estimates. The CoC is at the TCP.
- During the final stationary interval, mean differences are 2.32 percent for force and 4.87 percent for moment. The curves differ more during motion.

Baseline angular error

- With the CoC at the TCP, entry offsets of 0.69° , $+9.31^\circ$, and -9.41° gave angular errors of 1.65° , 1.75° , and 1.41° .
- The error magnitude decreased for both larger entry tilts, and the negative entry error changed sign. The nominal-zero condition increased from 0.69° to 1.65° .
- The bars show three-trial means, with one sample standard deviation. These are first-tangent components relative to the calibrated surface, calculated from recorded endpoints at 0.01° precision.

Effect of rotational stiffness

- Increasing rotational stiffness from 5 to 50 newton metres per radian increased the angular error from 1.75° to 6.58° . The stiffer rotational spring allowed less contact-induced correction.

Effect of CoC position

- The plot covers minus 80 to plus 80 millimetres, with three repetitions per setting. For positive entry tilt, the smallest tested angular error was 1.17° at plus 80 millimetres. At minus 80 millimetres, 9.35° remained.
- For negative entry tilt, the smallest angular error magnitude was 0.94° at plus 10 millimetres. At minus 80 millimetres, the error component crossed zero and ended at $+2.23^\circ$.
- CoC displacement changes the remaining error component and can reverse its direction. Greater rotation therefore does not always give a smaller angular error.

Angular error and interaction wrench

- In these traces, the error enters and stays within 0.1° of its final value after about 2.5 seconds at plus 40 millimetres. At the TCP this takes 3.6 seconds, about 1.1 seconds longer.
- Final estimated normal forces are similar, at about -83 , -79 , and -78 newtons.
- The interaction moment reverses with CoC displacement. These forces and moments come from the robot's model estimate.

Null-space controller

- The contact results are complete. The separate null-space study concerns the arm's redundant joint motion. At full Jacobian rank, seven joints leave one null-space direction with zero instantaneous EE motion.
- Damping reduces null-space motion by opposing redundant joint velocity.
- Conditioning chooses between two local null-space directions and drives towards the one with the larger minimum singular value. It can initiate joint motion. At a singularity, the EE loses an instantaneous motion direction. Near it, some EE motions require very large joint velocities. A larger minimum singular value indicates better local conditioning under the same Jacobian convention.

Null-space experiment

- The desired position and orientation stay fixed while joint torques apply a repeatable disturbance. This is equivalent to a virtual 20-newton point force on link three.
- We compare no null-space torque, damping, and two conditioning settings, with three trials each. Conditioning is active before and during the disturbance.
- We compare total projected joint motion, minimum singular value, and net projected displacement.

Cumulative null-space motion

- Cumulative motion, E_N , adds projected joint movement in both directions. These degrees describe arm motion, not rotation of the tool.
- Damping reduces it from 7.60° to 5.69° , about 25.1 percent. Conditioning gives 0.29° at $k_\sigma = 1.5$ and 1.69° at $k_\sigma = 2$.
- Curves show three-trial means over the displayed four-second interval. Shading shows one sample standard deviation.

Null-space displacement over time

- This plot retains movement direction. Motion back towards the start reduces the displacement.
- The seven projected joint velocities are integrated without taking their magnitude. Each trial uses the same reference direction as the net-displacement comparison. Positive motion follows the disturbance response without null-space torque.
- The left plot shows all four settings. The right plot enlarges the conditioning curves, in the same degree units. Both show three-trial means and one sample standard deviation.
- Conditioning produces movement away and back towards the starting value. These are projected joint displacements, not tool rotations.

Joint motion over time

- These curves show the measured angle of joint one relative to its value at disturbance onset. Joint one receives the largest equivalent disturbance torque in every trial.
- The left panel shows all four settings over four seconds. Each curve is an individual trial, with markers at the original samples taken at about 20 hertz.
- The right panel enlarges the first second for both conditioning settings. At the larger torque magnitude of 2 newton metres, all three trials show repeated motion reversals. The curves at 1.5 newton metres increase more smoothly.
- Subtracting the starting angle preserves both directions of joint movement. Conditioning was already active before the disturbance, so the settings can begin at different arm configurations.

Jacobian conditioning

- A smaller minimum singular value means some tool motions require larger joint velocities. Zero means the EE loses an instantaneous motion direction.

- It falls with no null-space torque and with damping. Both conditioning settings keep it nearly constant during the disturbance.
- Curves show three-trial means over the same four-second interval and use the same Jacobian convention. Shading shows one sample standard deviation.

Net null-space displacement

- Net displacement measures projected joint movement remaining after opposite motions cancel. It is 7.52° without null-space torque and 5.61° with damping.
- Mean conditioning displacements are only $1.5 \times 10^{-2}^\circ$ and $-1.1 \times 10^{-2}^\circ$. Conditioning counters the disturbance in the tested direction.
- Cumulative motions are 0.29° and 1.69° . The larger value is consistent with more back-and-forth motion despite similarly small net displacement.

Conclusion

- Contact reduced angular error for both larger entry tilts while the desired orientation stayed fixed. The error is relative to the calibrated surface and retains calibration and tool-mount uncertainty.
- Higher rotational stiffness increased angular error from 1.75° to 6.58° .
- CoC displacement changes angular error and can carry it through zero. At the TCP, the added moment vanishes while the rotational spring and damper remain active.
- Damping reduced arm motion, and conditioning countered the tested disturbance direction. Nearly zero net displacement coexisted with back-and-forth motion.
- Future work is a more rigid tool mount and adaptive CoC selection. After alignment, the adaptive CoC should return to the TCP.

Null-space torques and projector

- The kinematic projector selects null-space torque directions. It uses the Jacobian pseudoinverse J^+ .

$$N_\tau = I_7 - J^\top J^+$$

- Damping and conditioning add to give the total null-space torque.

$$\tau_{\text{null}} = \tau_d + \tau_\sigma$$

- Damping opposes projected joint velocity and is zero at rest. d_{null} is the damping coefficient.

$$\tau_d = -d_{\text{null}} N_\tau \dot{q}$$

- Conditioning probes both directions along the unit null-space vector v_7 . It chooses the direction with the larger minimum singular value.

$$\tau_\sigma = k_\sigma N_\tau (s_\sigma v_7)$$

- The selector s_σ is $+1$ or -1 , and zero within the deadband. k_σ sets the torque magnitude, which can be nonzero at rest.

Disturbance demonstration

- The controller holds the tool pose throughout the video. Without null-space torque, the disturbance moves the arm.
- Damping opposes the joint motion.
- Conditioning actively changes the arm configuration and can initiate motion. The tool stays approximately in place.